

Option A (my recommendation)

Train on:

2009-2017 + 2021-2022

Test on:

2023

or

Train: 2009-2017 + 2021

Test: 2022-2023

This is the best solution if labels exist for 2021/2022.

Why? --- Because then the model finally sees the new environment.

Option B

Keep:

Train : 2009-2017

Test : 2021-2023

but **don't use one model.**

Build two models.

Model 1:

2009-2017

Model 2:

2014-2017

Then average them.

Because:

- Model 1 captures rare bad events.
- Model 2 captures more recent behaviour.

Option C

Downsample the majority class.

The model learns: Everybody is good.

Use:

`scale_pos_weight = negatives / positives`

or

`class_weight="balanced"`

This won't fix drift, but it helps detect bads.

Option D

Train multiple windows and compare OOT.

Train 1: 2009-2017

Train 2: 2012-2017

Train 3: 2014-2017

Train 4: 2015-2017

Evaluate all on: 2021-2023

Then compare:

Train years OOT AUC

2009-2017 ?

2012-2017 ?

2014-2017 ?

2015-2017 ?

This is usually what banks do.

Train a two-model ensemble

Model 1:

2009-2017

Model 2:

2014-2017

Then:

```
final_proba = (  
    0.4 * proba_model1  
    +  
    0.6 * proba_model2  
)
```

This often works very well in credit risk.

Calibrate probabilities

Handle severe class imbalance